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Source: *Journal of Adolescent & Adult Literacy*, NOVEMBER/DECEMBER 2021, Vol. 65, No. 3 (NOVEMBER/DECEMBER 2021), pp. 227-236

Published by: International Literacy Association and Wiley

Stable URL: <https://www.jstor.org/stable/10.2307/48649629>

REFERENCES

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Checked Out? Videogame Programming in U.S. Libraries and Implications for Videogame Literacies

Sandra Schamroth Abrams, Richard N. Van Eck

Introduction

"Surprising patrons with our video game collections and programs. Many think of us as the library of the past and don't realize the services we offer. It's nice to meet patron interests/needs and change perceptions." (33-year-old female librarian, suburban public library, Idaho)

"It is a break from the normal teaching-oriented programming and it draws in the teens." (50-year-old male librarian, urban public library, Florida)

"It takes a lot of oversight to maintain control and flow." (28-year-old female librarian, rural public library, Michigan)

These librarians self-identify as women or men in their 20s to their 50s, who work in U.S. libraries in suburban, urban, or rural settings. And they all are talking about youth playing videogames in their respective libraries.

Although it might be surprising for some to think about videogames and associated practices as part of a library's programming, or planned events, we argue that it is a natural fit for libraries, which promote various literacies. After all, videogaming involves digital literacies, a multidimensional term with myriad definitions, as well as socio-technical, socio-economic, and socio-historical complexities and implications (Nichols & Stornaiuolo, 2019). Yet, librarians are not unified in their beliefs and practices regarding videogame programming, and little is known about the actual presence of videogame programming in libraries. How many libraries actually have videogame programming (i.e., formal and informal arrangements for videogame play)? What form does it take? What factors promote or constrain the presence of videogame programming? We set out to answer these questions to understand how such programming might support youth videogame literacies.

Videogames and Libraries: An Overview

For over 15 years, the inclusion of videogames in libraries has been the topic of single-authored and

edited compilations addressing the structure and role of game collections and game circulation, codes of conduct, library goals and curricula, and weekday play and tournament organization (Levine, 2006, 2008; Martin & Martinez, 2016; McNichol, 2009; Neiberger, 2007; Nicholson, 2013; Robson et al., 2017; Scordato & Forsyth, 2014). Neiberger (2007) offered librarians a manual for organizing and promoting two types of videogame programming: (1) open play, wherein youth select and play games at will, with or without specified play times and limits and (2) scheduled tournaments, or structured rounds of game play organized and managed by librarians and typically focused on one particular game.

Researchers have investigated topics related to videogame play and libraries, such as teacher-librarians' experiences with and perceptions of gaming (Hovious & Van Eck, 2015); youth-driven literacy practices and interaction during videogame play (Abrams, 2012; Abrams & Gerber, 2021; Adams, 2009; Martin & Martinez, 2016); and related civic engagement (Abrams, 2012; Brown & Kasper, 2013; Hollett, 2016; Hollett & Ehret, 2017).

Given that videogaming and its literacies are evident in the literature about library programming, it becomes important to investigate the factors supporting or constraining videogame programming—and, thus, its associated literacies—in libraries.

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Videogame Literacies

Videogame literacies involve a variety of digital and nondigital “meaning making that hinges on player recognition and understanding of various multimodal sign systems” (Abrams, 2016, p. 2319). Salen (2007) explained that gaming literacies include a player’s (a) knowledge of how a game works, (b) use of digital and nondigital resources, (c) ability to navigate socially, and (d) involvement in collaboration and knowledge sharing. Relatedly, interaction distinguishes videogames from other media (Greenfield, 1984), and videogame literacies include interaction not only with the videogame, but also with the gaming space and the players and spectators central and peripheral to videogame play.

Research also suggests that “gaming is thoroughly a literacy practice, requiring players to produce meaning with texts and become expressive with technology in multiple forms” (Squire, 2008, p. 636). Although videogame play might involve processes, strategies, and engagement similar to reading in general (Beavis, 1998), the literacy practices related to videogaming require an understanding of *literacies* (in its plural form)—a well-established concept that extends beyond print to include nuanced yet expansive meaning making (New London Group, 1996; Street, 1984, 1999).

This article addresses videogame literacies to underscore the participatory meaning making in and around videogaming, practices that involve iterative learning often via trial-and-error approaches and peer feedback (cf. Abrams, 2009, 2015; Abrams & Gerber, 2013, 2021; Gee, 2003; Hanghøj, in press; Squire, 2010, 2011; Steinkuehler et al., 2012; Van Eck, 2008).

So What is the Issue?

Videogame play engages learners (cf. Annetta, 2008; Bacalja, in press; Clark et al., 2016; Gee, 2003; Hanghøj, in press; Hawisher & Selfe, 2007; Schaffer et al., 2005); it provides an experience-based context for academic material (cf. Abrams, 2009; Barab et al., 2009; Gerber & Price, 2011; Squire, 2011; Wainwright, 2014); and it helps students to achieve higher-order problem solving (Van Eck & Dempsey, 2002). Despite these benefits, videogames are not widely adopted in schools, and research (e.g., Bell & Gresalfi, 2017) suggests that teachers’ experiences with videogames are one reason why. Research reveals that common barriers to the uptake of videogames and game elements include lack of training and resources, lack of interest, quality and user-friendliness of hardware and software, and overall impact on content (i.e., fit) and classroom management (cf. Kaimara et al., 2021; Sánchez-Mena & Martí-Parreño, 2017).

Might these barriers also be true for librarians, who historically have supported student literacy and learning? Despite research of videogaming in libraries (Levine, 2006, 2008; Martin & Martinez, 2016; McNichol, 2009; Neiberger, 2007; Nicholson, 2013; Robson et al., 2017; Scordato & Forsyth, 2014), little currently is known about videogame literacies in libraries and reasons explaining the degree to which such programming (i.e., formal or informal activities or events wherein youth play videogames in the library) occurs.

Thus, our study draws upon librarian feedback regarding the presence of videogame programming, the adolescents’ interaction in a videogame space, and the communicative and participatory interplay of videogame literacies in the library. If videogames are not widely present, then what are the reasons for this? When it exists, what does videogame programming look like in libraries?

Insight into library videogame programming is central to understanding how these spaces support youth literacies through engagement in meaning-making experiences that are steeped in social interaction and multi-textual practices. Likewise, understanding how librarians think about and experience videogame programming might help us better understand these literacy practices and the factors that support or impede them. Such questions are not limited to libraries, and we hope that our research will help educators, researchers, librarians, and anyone else interested in youth literacies to understand better the conditions that might promote or inhibit videogame literacies in informal settings.

Videogame Programming: Supporting and Constraining Factors

There are external and internal factors that support or inhibit the creation of library videogame programming. To understand these factors, we adopted the framework of first- and second-order barriers, initially proposed by Fullan & Stiegelbauer (1991) to explain barriers to educational change and later adapted by Ertmer (1999) to address barriers to technology integration. First-order barriers are external (e.g., systems, processes) and second-order barriers are internal (e.g., beliefs). Barriers are a more precise way of measuring attitudes toward technology because barriers can be used to classify positive and negative attitudes according to whether they are external or internal and, thus, require different interventions.

First-Order Barriers

First-order, or extrinsic, barriers refer to a lack of resources that librarians have access to at a given point

in time (e.g., funding, hardware or software, Internet access, physical space, and furniture to support videogame play). Addressing first-order barriers is relatively straightforward (although not necessarily simple) and usually requires additional resources, professional development, and changes to curriculum, accreditation practices, and/or policy.

Second-Order Barriers

Second-order, or intrinsic, barriers also can impact the inclusion of videogame programming at a library. These barriers are rooted in librarians' knowledge and experiences, as well as belief systems and perceptions, such as whether videogames have value, are too complex, take too long, and will promote rowdy behavior in the library. Addressing these barriers might involve some of the same solutions as first-order barriers, but an individual's intrinsic characteristics might be very difficult to change, and modifications (e.g., personnel selection, recruitment, and promotion criteria) might be necessary.

In order to advance our understanding of videogame literacies in libraries, it is crucial to determine the extent to which videogame programming even exists, what form it takes, and what kind of barriers explain its presence and connection to videogame literacy practices.

Methods

This article draws upon data from a larger, integrated mixed methods study of more than 1,000 U.S. librarians from all 50 states (Abrams & Van Eck, in press). We sought equal-status mixed research (Johnson et al., 2007) by giving "equal prominence" to the quantitative and qualitative methods and data (Mayoh & Onwuegbuzie, 2015, p. 98). This article features a portion of the quantitative and qualitative data derived from the online Qualtrics survey administered between August 2017 and May 2018 and analyzed between 2018 and 2019.

Survey Design

The survey included 26 open-ended and 21 multiple-choice and Likert-type questions offering a snapshot of (a) locations, settings, programming, roles, and responsibilities; (b) the presence and characteristics of videogame programming (librarians without videogame programming responded to the same questions regarding hypothetical programming); and (c) videogame play, device ownership, and barriers to the adoption of

videogame programming. In addition, answers to the open-ended questions provide insight into perceived positive and negative aspects of videogame programming and videogame literacies (see Appendix 1).

First- and second-order barriers were assessed using two subscales of the Teachers' Attitudes Toward Games, or TATG (Hovious & Van Eck, 2015), an 81-item instrument developed by classifying items and outcomes from the 15 most cited teacher attitude toward games instruments according to Ertmer's framework. The subscales were created based on clustering those items by similarity of focus. Gaming Difficulty (an 8-item, first-order subscale) measured whether librarians felt that videogames were too complex a technology to integrate in libraries with existing training, support, and resources. Gaming Confidence (a 9-item, second-order subscale) measured the extent to which librarians believed they, personally, would be able to work with videogames (e.g., if programming were to be offered in their library).

Participants

Participants included librarians in public, private, academic, and special libraries. Differences in settings were not examined because the primary focus of the study was on the relationship of barriers to the presence and nature of videogame programming.

We began by contacting members of the American Library Association (ALA), Academic and Special Libraries faculty, and colleagues who engage in library research. We also joined 14 librarian email groups and emailed or called representatives from library organizations from all 50 states (including the District of Columbia), sharing the survey and asking that they share the survey as well. At least eight librarians noted that they distributed the survey to their colleagues in and beyond their libraries. We also joined email groups and contacted specific people as recommended by librarians. Thus, the snowball sampling method (Goodman, 1961) helped to support initial data collection efforts.

Data Collection

In October 2017, we examined the responses per state in relation to each state's population percentage relative to the United States and initiated another round of email solicitation, specifically targeting states that seemed underrepresented. In May 2018, with over 1,300 combined national and international responses and a noticeable lull in participation, we determined that each state was represented in proportion to its population.

A total of 1,016 U.S. librarians responded to the survey. Because many responses were optional and/or enabled respondents to select more than one option, percentages and total responses per question presented in the following sections will differ from each other and from the total number of survey respondents. Respondents came from every state in the United States and in proportions roughly equivalent to each state's population. Approximately 79.3% (806) self-identified as women, 17.8% (181) self-identified as men, and 2.9% (29) did not self-identify. Of those who responded to the question regarding library setting (5% left it unanswered), 26% indicated an urban setting, 38% indicated a suburban setting, 28% indicated a rural setting, and 3% indicated "Other." The majority of respondents (726, or 71.5%) reported holding a Library Science graduate degree and had worked in their respective libraries for approximately 12 years. Librarians reported a wide range of roles held currently and previously, the majority of which were in public libraries or university libraries.

Data Analysis

Quantitative analysis for this phase of the study was accomplished using SPSS software version 25. Data were screened for completeness and accuracy before running a variety of analyses, including frequencies and descriptives for demographic and multiple-choice questions regarding presence of programming, and Pearson product-moment binary correlations to test relationships among responses.

Positively worded items (e.g., "I don't think games are difficult at all") on the Gaming Difficulty (first-order barriers) and Gaming Confidence (second-order barriers) subscales, where agreement on the Likert-type scales would indicate the *absence* of barriers, were recoded and averages were created for each subscale. Thus, "Strongly Agree," which was a higher number, indicated a higher *presence* of barriers.

Qualitative analysis was accomplished using ATLAS.ti software 8.2.33, which supports coding and theming (Saldaña, 2013). Qualitative data provided insight into what librarians liked most and liked least about videogame programming, which also included explicit and implicit notions about literacies, in general, and videogaming literacies, in particular. Where appropriate, these data were interpreted within the constructs of first-order (Gaming Difficulty) and second-order (Gaming Confidence) barriers. Initial codes included behavior, management/oversight, acceptance (or lack thereof), patronage, funding, space, enjoyment, and

community. Such data analysis helped to clarify *why* first- and/or second-order barriers existed and how programming informed videogame literacies (and vice versa).

Prevalence of Library Videogame Programming

Literature on videogaming in libraries makes the practice seem pervasive, yet we found that the majority of librarians did *not* offer videogaming as an option in their library: 384 (48%) indicated that they had one and 420 (52%) indicated that they did not.

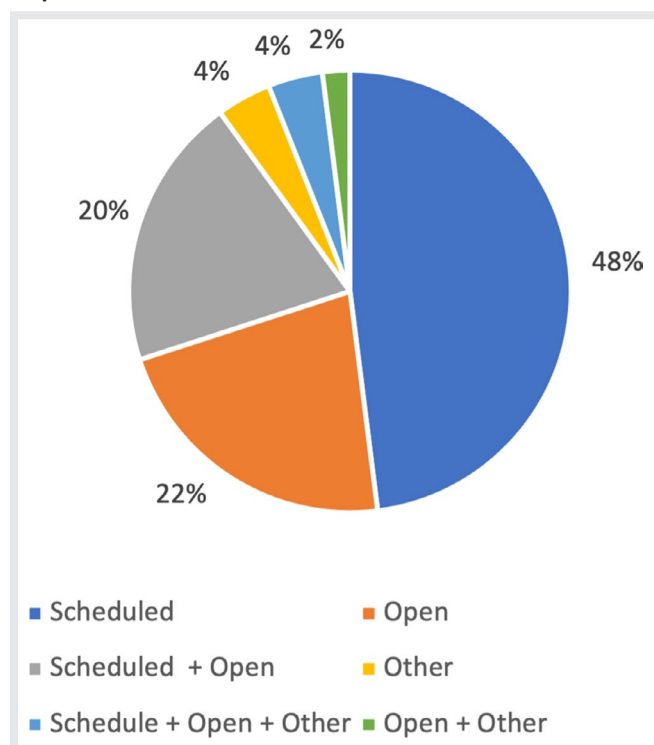
Despite this relatively low prevalence, our data suggest evidence of growth and progress. For example, Nicholson's (2009) study revealed that fewer than 50% of libraries had some kind of formal game programming, and of those, only 13% focused on *videogaming* specifically. Although our study confirms that fewer than 50% of libraries had videogame programming, our data reveal that, of the ones that did, most (63%) have had videogame programming for less than 5 years, with 35% only in existence less than 3 years. Such data suggest that videogame programming is a relatively new phenomenon at almost all (98%) of the libraries that featured such programming. One librarian, noting the underdeveloped nature of the videogame programming at the library, exclaimed, "at least it's here!!! the library I now (4 months tenure) work for is not quite as progressive in its offering as it could be." All these data work in concert to illustrate the importance of understanding the relationship between videogame programming and librarians' perceptions.

Types of Programming and (Limited) Practices

Of the 384 librarians who reported some form of videogame programming, almost half (48%) relied solely on scheduled access (play allowed on specific days and times), about one-fifth (22%) relied on open play (play allowed any time), one-fifth (20%) reported only offering both forms of programming, and 4% reported only some other form of programming (e.g., games and/or consoles for checkout). Figure 1 provides a detailed breakdown of videogame programming types.

Scheduled videogaming follows the organized nature of traditional library programming (e.g., certain programming runs on particular days), but might suggest limitations in librarians' understanding of videogame literacies. Because videogame literacies often are developed through the ongoing, iterative nature of free

Figure 1
Percentages of Videogame Programming Types Reported



Note. The color figure can be viewed in the online version of this article at <http://ila.onlinelibrary.wiley.com>.

play, scheduled programming is restrictive. Additionally, patrons without easy access to private or public transportation might not be able to attend during the specified times. Scheduled programming comprised almost half of all programming, representing a significant mismatch of programming and videogame literacies.

Conversely, 22% of libraries with videogame programming offered *only* open programming, suggesting a potential missed opportunity to extend play in specific ways (e.g., tournaments, group play). It is encouraging that 24% of libraries offered at least both scheduled and open play, thus maximizing the potential impact of videogame programming; it will be critical to understand how to promote this model more widely in libraries.

In many cases (54%), there were time limits on videogame play, and 41% of librarians noted that adolescents played the games they, the librarians, selected. However, 72% of librarians also indicated that patrons could choose their games to play during scheduled and/or unscheduled times, suggesting that librarians were able to adopt a youth-interest approach to promoting literacy rather than relying only on specific titles and designed experiences.

Videogame Programming and Literacy Experiences: Supports or Barriers

Recognizing the first- and second-order barriers that impact library videogame programming is an important first step to understand how, if at all, youth might be engaging in and honing videogame literacies in informal spaces, such as libraries, and to articulate the steps that can be taken to remove barriers to videogame literacies. In what follows, we discuss the data that reveal extrinsic and intrinsic factors influencing librarians' desire (or lack thereof) to embrace videogame programming.

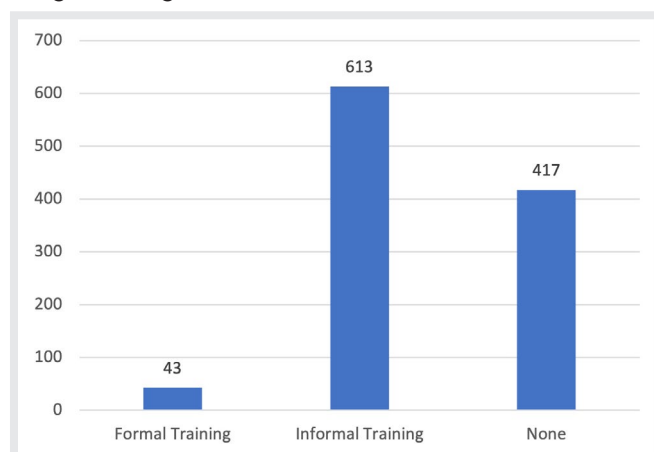
First-Order Barriers

Resources and Support. First-order barriers were measured in part by the 8-item TATG subscale "Gaming Difficulty." Overall, librarians tended to "agree" (3.71 out of 5) that the difficulty of supporting videogames was a significant barrier to programming. Whereas most respondents felt it would be "somewhat easy" ($M = 3.35$ out of 5) to secure funding for new programming they envisioned, they reported that it was "somewhat difficult" ($M = 2.40$) to get new programming funded overall.

Unfortunately, librarians saw this as an uphill battle when it came to videogame programming, specifically. When asked how likely it would be to run videogame programming based on existing or new funds, 187 librarians (45%) indicated why the current library structure would not support such an effort (e.g., "My library would not value that kind of program" and "Our funds are limited and most of our programming is done on a shoestring").

Training. When it came to professional development, librarians most frequently selected "Informal Training" as the type of training they received on the design and implementation of videogame programming (see Figure 2). Further analysis revealed that those informal sources primarily stemmed from independent searches (40%), informal conversation (50%), and feedback from other librarians (44%). Surprisingly, Library Science degrees accounted for less than 12% of training, and formal professional development accounted for 22%. This suggests that, beyond the personal impetus for learning how to start and develop videogame programming in the library, librarians have few resources or external supports, and university programs might not be addressing this problem sufficiently. Librarians tended to list training needs in the same breath as resources and funding: "to do a videogaming literacy program, in addition to the

Figure 2
Self-Reported Sources of Training for Videogame Programming



Note. Respondents could select more than one source. The color figure can be viewed in the online version of this article at <http://ila.onlinelibrary.wiley.com>.

funds, I would need extra training to learn how to make this happen. Training and staffing would be the biggest hurdles."

If librarians are learning about videogame programming in ad hoc ways, then it makes them even less likely to be successful in advocating for such programming and related youth literacies. It will be crucial to promote educational reform of Library Science programs as a first step in providing this training.

Second-Order Barriers

Second-order barriers were measured in part by the 9-item "Gaming Confidence" subscale of the TATG. Librarians tended to "agree" that they were not confident in their own abilities to play videogames and support videogame play (3.76 on a 5-point scale). While one might assume that librarians did not have much exposure to games, survey responses suggest otherwise.

Seven hundred, or 69%, of survey respondents reported personal ownership of at least one gaming device, with an average ownership of three devices. Common devices were computers (40%), and the Wii (40%), Xbox (28%), and PlayStation (26%) platforms. Of the 691 librarians who provided a self-reported stance on videogames (see Figure 3), the majority (418 or 60%) had a positive perspective about videogaming. However, librarians did not play videogames regularly—only 25% reported playing a videogame in the last month and on an average of 3.6 hours per week at home. Approximately half (333 of 677 responses) indicated they do not play at all during the week.

Interactivity and Librarian Videogame Literacy Beliefs

Relatedly, barriers to videogame programming in libraries include the beliefs of librarians. To implement videogame programming, librarians need to be open to the literacies videogaming practices involve. Although our initial survey questions focused on programming and practices (i.e., we did not explicitly use the words, literacy or literacies), the data included discussions of literacy; one 65-year-old female librarian from Louisiana explained that "we tie nearly all programming of any kind to literacy."

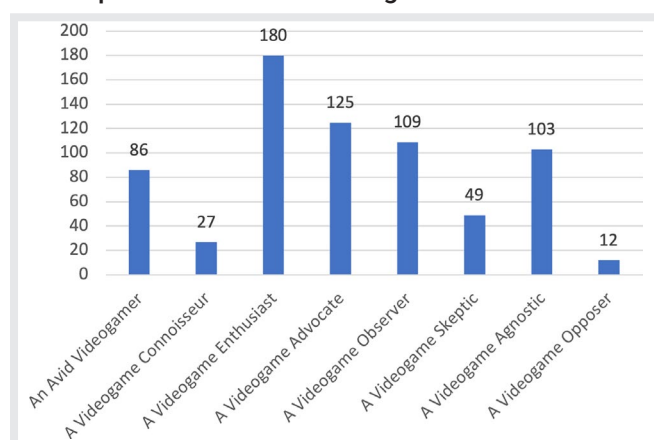
Even if the words, "literacy" or "literacies," were not specifically verbalized, some librarians spoke about the value of social interaction in videogame meaning-making practices, which clearly is tied to videogame literacies. In all, 32 librarians spoke positively about the community effort their videogame programming fostered. A 45-year-old female New York librarian enjoyed "seeing the teens bond," and a 34-year-old male librarian from Missouri appreciated "seeing young people enjoying themselves and seeing the library as a cool place to be." Others, including a 31-year-old female librarian from Virginia, recognized the importance of "creative play and interaction of the kids playing."

When explaining why they like the videogame programming at their libraries, librarians noted adolescent literacies that included, but also extended beyond, language. One 34-year-old female librarian from Delaware particularly enjoyed witnessing "The passion of the program that uses it - Japanese language/culture." And a 27-year-old female Maine librarian spoke about "seeing cooperative play and problem-solving between teens who might not normally interact."

Furthermore, librarians specifically acknowledged and valued the communication, problem solving, and interaction that videogame programming supports. In fact, 15 librarians even focused on *their* interaction with teens, as well as intergenerational gaming. These librarians specifically enjoyed speaking and connecting with teens: "Getting to know the teens and having fun teaching them the game and playing with them." One of the librarians noted her role in "thinking [with the teens] about the business of the gaming industry, talking with teens about my love of games. Allowing them to see a female play games and enjoy them."

These instances of shared literacies and meaning making also extended to adult family members. Despite "complaints from adult patrons," six librarians specifically appreciated how teens first work together "incorporating designing of games, testing of games" and then play with adult family members via "intergenerational gameplay."

Figure 3
Self-Reported Stance on Videogames



Note. The color figure can be viewed in the online version of this article at <http://ila.onlinelibrary.wiley.com>.

Likes and Dislikes

We also asked librarians questions about their favorite and least favorite part of the videogame programming at their library. There were 359 responses about a favorite component of the videogame programming and 335 responses describing least favorite components. Of the librarians who detailed their favorite aspects of videogame programming, almost half (165, or 46%) seemed interested in videogaming as a social practice wherein youth “have fun” and interact with each other and the game. Two female librarians in their mid-30s noted that videogame programming at their respective Oregon and Wyoming libraries created a specific space for youth and “socialization opportunities for teens who may otherwise be isolated at home.”

Despite the excitement some librarians voiced, 109 bemoaned management and oversight difficulties, including issues of resources and support (51/109). As one 34-year-old female librarian from Pennsylvania explained, the videogame programming meant “Losing 2 hours *every* week out of my schedule to supervise the program.” Additionally, a 33-year-old female librarian from Ohio elaborated on the time and effort it takes to organize and prepare for the videogame programming:

Setting everything up. We have so many different people working with our consoles that everything is always in disarray. On top of that, people are always changing the A/V settings in our meeting room so there’s usually at least an hour of trying to puzzle things out before everything is up and running correctly.

Setting up videogame equipment requires literacy practices to “puzzle things out”—from understanding A/V settings to knowing what controllers and related practices a particular videogame involves. This might help explain the gaming difficulty ratings on the subscale of the TATG.

Additionally, 78 librarians made a special note of adolescent behavior and hygiene, such as noise, theft, and body odor. One 38-year-old librarian from Kentucky explained that her least favorite part is “what the room smells like after hours of middle schoolers + Doritos + pizza,” which is similar to what a 54-year-old female librarian from Indiana said about the “teenage boy smell and eating habits.”

Then there is the issue of buy-in because adult patrons or librarians perceived adolescents’ game play and the videogame programming to lack value. A 30-year-old librarian from Washington state spoke of patron “comments and complaints...about the noise level or appropriateness for teens and children to be playing games in the library.” And a 54-year-old female librarian from Montana spoke of literal buy-in: “Videogaming has never been strongly supported by administration, so it is poorly funded,” a sentiment echoed by a 31-year-old librarian from Ohio, who said that “acceptance from other library staff” was her least favorite part of the videogame programming. Furthermore, a 61-year-old female librarian from New York thought that the videogame programming “takes up valuable space that we could use for other things,” and a 49-year-old female librarian from New York spoke of how programming “gets the teens out of my hair for a bit.”

Interpretation

Videogaming involves iterative meaning making with multimodal spaces and sign systems, and the participatory literacies—from collaborative feedback to iterative discovery and strategizing—have implications in and beyond academic contexts (cf. Abrams, 2009, 2015; Gerber & Price, 2011; Hanghøj, in press). As informal learning spaces deeply connected to the communities in which they are located, libraries—and by extension, librarians—play a critical part in promoting literacy, in general, and potentially with videogame literacies, specifically. Our findings on the first- and second-order barriers can provide insight into the videogame programming available to youth, the literacies promoted or endorsed in U.S. libraries, and how youth might be honing videogame literacies in informal spaces, such as libraries.

Librarians are not equally prepared and motivated to promote videogame literacies. The adoption and diffusion of videogame practices in libraries is profoundly

influenced by the attitudes, knowledge, and training of the people who are responsible for promoting videogame practices (the second-order barriers), as well as the resources and constraints of the environments in which they work (the first-order barriers).

Recognizing the factors that contribute to the support (or lack thereof) of library videogame programming is an important first step in promoting the creation and adoption of such programming and the related literacies inherent in and around videogame play. Our research calls attention to how librarians' experiences with and beliefs about such programming might influence the availability and success of videogaming and related literacies in the library.

Implications

Our findings are applicable not only to librarians in public, school, and academic libraries, but also to educators, in general, and to English Language Arts teachers, in particular, who promote the development of students' literacies (cf., Farmer & Murphy, 2010; Hovious & Van Eck, 2015).

Although beyond the scope of this article, it will be important to address any inequities related to library videogame programming and practices across urban, suburban, and rural settings, as well as the implications for youth and adult literacies. Additionally, while we address the data primarily in terms of first- and second-order barriers, it is important to note that these barriers can be interrelated and that it might not be possible to address one without the other. Attitudes and beliefs, such as those that result from second-order barriers, are the result of multifaceted, complex interaction among many variables, including first-order barriers, stereotypical perceptions of youth and videogames, or some interaction between the two. The gendered and stereotypical nature of some of the librarian comments, for example, are a good reminder to avoid assuming causality or correlation among the data sources, to be mindful that videogame behavior is contingent on many variables not explored in this research, and to consider carefully the first- and second-order barriers.

First-Order Barriers

The most significant first-order barriers identified were resources and support, training, and funding. To overcome these barriers, we recommend greater focus on librarian/educator professional development activities that include videogame literacies and their connection to youth meaning making. This extends to library science as well to pre-service teacher-education programs.

Relatedly, we suggest library accreditation bodies modify their standards for digital literacy to encompass videogaming as it relates to programming, resources, and practices. We also recommend starting small—a videogame discussion club (cf. Stuftt et al., 2016) akin to the virtual book clubs that many libraries have employed during the pandemic, can be an effective way to test the waters.

Second-Order Barriers

The second-order barriers, which are intrinsic and thus the most challenging to address, included librarians' experience with videogames, their perceived value of videogaming and teen interaction, and their role in managing and overseeing videogame programming. Modifications to library science and teacher education programs and their accreditation standards could support a shift in practice and culture for graduates who share the beliefs about videogaming as legitimate practice. Likewise, professional development might help to mitigate second-order barriers for current and future librarians. Given that dislikes tended to focus on management of programming, we also recommend the reorganization of youth library programming to accommodate adolescents' and librarians' needs. Youth participation in the development of library videogame programming (Abrams & Gerber, 2021), along with sustained professional development, will help librarians reinvent programming and related curriculum development.

Final Thoughts

Because videogaming is a prevalent activity among teens (Lenhart et al., 2015) and because one of the goals of libraries is to promote literacies through programming that will attract the public, including teenagers, there is a widespread perception that videogame programming will be common. Our research suggests that such programming is neither ubiquitous nor universal. Nonetheless, videogame programming can be "effective community builders and catalysts for increased awareness and usage of library sources and services" (Boyle, 2018, p. 2), also providing youth access to gaming platforms, software, and practices.

Through videogame play in libraries, adolescents interact with multiple, multimodal texts in a variety of ways; meaning making in these spaces "tends to be a social experience with friends and family, giving children and youth the opportunity to learn by observing the behavior of these individuals" (Mills et al., 2017, p. 28). Youth who are present in library videogame spaces as players

or active spectators also often layer their literacies as they move in, around, and about various digital and non-digital practices (Abrams, 2015; Abrams & Russo, 2015). Librarians, therefore, are uniquely suited to promote literacy and learning through videogame programming (Farmer & Murphy, 2010; Hovious & Van Eck, 2015).

Recognizing the state of play in U.S. libraries, as well as the first- and second-order barriers impacting videogame programming, is an important step to understand the factors influencing the support and/or development of youth literacies, in general, and youth videogame literacies, in particular, in informal spaces like libraries.

Conflict of Interest

Sandra Schamroth Abrams and Richard Van Eck have no conflicts of interest.

REFERENCES

- Abrams, S. S. (2009). A gaming frame of mind: Digital contexts and academic implications. *Educational Media International*, 46(4), 335–347. <https://doi.org/10.1080/09523980903387480>
- Abrams, S. S. (2012). Powerful gaming structures and practices: Videogames, situated language, and cultural contexts. *Languages and Linguistics*, 30, 41–63.
- Abrams, S. S. (2015). *Integrating virtual and traditional learning in 6–12 classrooms: A layered literacies approach to multimodal meaning making*. Routledge.
- Abrams, S. S. (2016). Videogaming and literacies. In M. Peters (Ed.), *Encyclopedia of educational philosophy and theory* (pp. 2318–2323). Springer.
- Abrams, S. S., & Gerber, H. R. (2013). Achieving through the feedback loop: How videogames can help us understand authentic assessment and meaningful learning. *English Journal*, 103(1), 95–103. <https://www.jstor.org/stable/24484067>
- Abrams, S. S., & Gerber, H. R. (2021). *Videogames, libraries, and the feedback loop: Learning beyond the stacks*. Emerald.
- Abrams, S. S., & Russo, M. P. (2015). Layering literacies and contemporary learning. *Journal of Adolescent & Adult Literacy*, 59(2), 131–135. <https://doi.org/10.1002/jaal.447>
- Abrams, S. S., & Van Eck, R. N. (in press). A Tale of inherent integration: Process methodology in support of a shared purpose to investigate videogame programming in libraries. In A. J. Onwuegbuzie & J. Hitchcock (Eds.), *Routledge handbook for advancing integration in mixed methods research*. Routledge.
- Adams, S. S. (2009). The case for video games in libraries. *Library Review*, 58(3), 196–202. <https://doi.org/10.1108/00242530910942045>
- Annetta, L. A. (2008). Video games in education: Why they should be used and how they are being used. *Theory Into Practice*, 47(3), 229–239. <https://doi.org/10.1080/00405840802153940>
- Bacalja, A. (in press). A critical review of digital game literacies in the English classroom. *L1-Educational Studies in Language and Literature*.
- Barab, S. A., Scott, B., Siyahhan, S., Goldston, R., Ingram-Goble, A., Zuiker, S. J., & Warren, S. (2009). Transformational play as a curricular scaffold: Using videogames to support science education. *Journal of Science Education and Technology*, 18(4), 305–320. <https://doi.org/10.1007/s10956-009-9171-5>
- Beavis, C. (1998). Computer games, culture and culture. In I. Snyder (Ed.), *Page to screen: Taking literacy into the electronic era* (pp. 234–255). Routledge.
- Bell, A., & Gresalfi, M. (2017). Teaching with videogames: How experience impacts classroom integration. *Technology, Knowledge, and Learning*, 22(3), 513–526. <https://doi.org/10.1007/s10758-017-9306-3>
- Boyle, C. (2018). Play time: Why video games are essential to urban academic libraries. *Urban Library Journal*, 24(1), 1–11. <https://academicworks.cuny.edu/ulj/vol24/iss1/3>
- Brown, R. T., & Kasper, T. (2013). The fusion of literacy and games: A case study in assessing the goals of a library video game program. *Library Trends*, 61(4), 755–778. <https://doi.org/10.1353/lib.2013.0012>
- Clark, D. B., Tanner-Smith, E. E., & Killingsworth, S. S. (2016). Digital games, design, and learning: A systematic review and meta-analysis. *Review of Educational Research*, 86(1), 79–122. <https://doi.org/10.3102/0034654315582065>
- Ertmer, P. A. (1999). Addressing first- and second-order barriers to change: Strategies for technology integration. *Education Technology, Research and Development*, 47(4), 47–61. <https://doi.org/10.1007/BF02299597>
- Farmer, L. S., & Murphy, N. G. (2010). eGaming and girls: Optimizing use in school libraries. In R. Van Eck (Ed.), *Interdisciplinary models and tools for serious games* (pp. 306–332). Information Science Reference.
- Fullan, M., & Stiegelbauer, S. M. (1991). *The new meaning of educational change*. Teachers College Press.
- Gee, J. P. (2003). *What video games have to teach us about learning and literacy*. Palgrave MacMillan.
- Gerber, H. R., & Price, D. (2011). 21st century learners, writing, and new media: Meeting the challenge with game controllers. *English Journal*, 2(101), 68–73. <https://www.jstor.org/stable/41415429>
- Goodman, L. A. (1961). Snowball sampling. *Annals of Mathematical Statistics*, 32(1), 148–170. <https://doi.org/10.1214/aoms/1177705148>
- Greenfield, P. M. (1984). *Mind and media: The effects of television, video games, and computers*. Harvard University Press.
- Hanghøj, T. (in press). Teachers' framing and dialogic authoring of Minecraft in the L1 classroom. *L1-Educational Studies in Language and Literature*.
- Hawisher, G., & Selfe, C. (Eds.). (2007). *Gaming lives in the twenty-first century*. Palgrave Macmillan.
- Hollett, T. (2016). Interests-in-motion in an informal, media-rich learning setting. *Digital Culture & Education*, 8(1), 1–19.
- Hollett, T., & Ehret, C. (2017). Civic rhythms in an informal, media-rich learning program. *Learning, Media and Technology*, 42(4), 483–499. <https://doi.org/10.1080/17439884.2016.1182926>
- Hovious, A., & Van Eck, R. N. (2015). Digital games for 21st-century learning: Teacher librarians' beliefs and practices. *Teacher Librarian*, 42(5), 34–38.
- Johnson, R. B., Onwuegbuzie, A. J., & Turner, L. A. (2007). Toward a definition of mixed methods research. *Journal of Mixed Methods Research*, 1(2), 112–133. <https://doi.org/10.1177/1558689806298224>
- Kaimara, P., Fokides, E., Oikonomou, A., & Deliyannis, I. (2021). Potential barriers to the implementation of digital game-based learning in the classroom: Pre-service teachers' views.

- Technology, Knowledge and Learning*. <https://doi.org/10.1007/s10758-021-09512-7>
- Lenhart, A., Smith, A., Anderson, M., Duggan, M., & Perrin, A. (2015). *Teens, technology, & friendships: Video games, social media and mobile phones play an integral role in how teens meet and interact with friends*. Pew Research Center. <http://www.pewinternet.org/2015/08/06/teens-technology-and-friendships/>
- Levine, J. (2006). Gaming and libraries: Intersection of services. *Library Technology Reports*, 42(5), 5–67.
- Levine, J. (2008). Gaming and libraries update: Broadening the intersections. *Library Technology Reports*, 44(3), 1–42.
- Martin, C., & Martinez, R. (2016). Games in classroom and practice in library and information science education. *On the Horizon*, 24(1), 82–87. <https://doi.org/10.1108/OTH-08-2015-0051>
- Mayoh, J., & Onwuegbuzie, A. J. (2015). Toward a conceptualization of mixed methods phenomenological research. *Journal of Mixed Methods Research*, 9(1), 91–107. <https://doi.org/10.1177/1558689813505358>
- McNichol, S. (Ed.). (2009). Web 2.0 and gaming in libraries. *Library Review*, 58(3), 167–251.
- Mills, J. E., Campana, K., & Goldsmith, A. Y. (2017). Libraries as learning labs in a digital age: A youth services conference in an LIS classroom. *Journal of Education for Library and Information Science*, 58(1), 27–38. <https://doi.org/10.3138/jelis.58.1.27>
- Neiburger, E. (2007). *Gamers ... in the library?! The why, what, and how of videogame tournaments for all ages*. American Library Association.
- New London Group. (1996). A pedagogy of multiliteracies: Designing social futures. *Harvard Educational Review*, 66, 60–92.
- Nichols, T. P., & Stornaiuolo, A. (2019). Assembling “Digital Literacies”: Contingent pasts, possible futures. *Media and Communication*, 7(2), 14–24. <https://doi.org/10.17645/mac.v7i2.1946>
- Nicholson, S. (2009). Go back to start: Gathering baseline data about gaming in libraries. *Library Review*, 58(3), 203–214. <https://doi.org/10.1108/00242530910942054>
- Nicholson, S. (2013). Playing in the past: A history of games, toys, and puzzles in North American libraries. *The Library Quarterly*, 83(4), 341–361. <https://doi.org/10.1086/671913>
- Robson, D., Parks, S., & Miller, E. D. (2017). Building game collections in academic libraries: A case study at the University of North Texas. In M. Robison & L. Shedd (Eds.), *Audio recorders to zucchini seeds: Building a library of things* (pp. 171–186). Libraries Unlimited.
- Saldaña, J. (2013). *The coding manual for qualitative researchers*. SAGE.
- Salen, K. (2007). Gaming literacies: A game design study in action. *Journal of Educational Multimedia and Hypermedia*, 16(3), 301–322.
- Sánchez-Mena, A., & Martí-Parreño, J. (2017). Drivers and barriers to adopting gamification: Teachers’ perspectives. *The Electronic Journal of E-Learning*, 15(5), 434–443.
- Schaffer, D., Squire, K., Halverson, R., & Gee, J. (2005). Video games and the future of learning. *Phi Delta Kappan*, 87(2), 105–111.
- Scordato, J., & Forsyth, E. (2014). *Teen games rule!: A librarian’s guide to platforms and programs*. Libraries Unlimited.
- Squire, K. D. (2008). Video-game literacy: A literacy of expertise. In J. Coiro, M. Knobel, C. Lankshear, & D. J. Leu (Eds.), *Handbook of research on new literacies* (pp. 635–670). Routledge.
- Squire, K. D. (2010). From information to experience: Place-based augmented reality games as a model for learning in a globally networked society. *Teachers College Record*, 112(10), 2565–2602.
- Squire, K. (2011). *Video games and learning: Teaching and participatory culture in the digital age*. Teachers College Press.
- Steinkuehler, C., Squire, K., & Barab, S. (Eds.). (2012). *Games, learning, and society: Learning and meaning in the digital age*. Cambridge University Press.
- Street, B. V. (1984). *Literacy in theory and practice*. Cambridge University Press.
- Street, B. V. (1999). Literacy and social change: The significance of social context in the development of literacy programmes. In D. A. Wagner (Ed.), *Future of literacy in a changing world* (pp. 55–72). Hampton Press.
- Stuft, C., Abrams, S. S., & Gerber, H. R. (2016). Critical thinking and layered understandings: Book clubs, videogames, and adolescent learning. *The ALAN Review*, 43(3), 96–102.
- Van Eck, R. (2008). COTS in the classroom: A teacher’s guide to integrating commercial off-the-shelf (COTS) games. In R. Ferdig (Ed.), *Handbook of research on effective electronic gaming in education* (pp. 179–199). Idea Group.
- Van Eck, R., & Dempsey, J. (2002). The effect of competition and contextualized advisement on the transfer of mathematics skills in a computer-based instructional simulation game. *Educational Technology Research and Development*, 50(3), 23–41.
- Wainwright, A. M. (2014). Teaching historical theory through video games. *History Teacher*, 47(4), 579–612.

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Survey of Videogames in Libraries.